SIMATS ENGINEERING
ASSESSMENT TOOL 1

COURSE NAME: Computer Vision with OpenCV COURSE CODE: DSA02

COURSE OUTCOME COVERED

CO1: *Apply the fundamental concepts of image formation, pinhole to perspective camera models, homogeneous coordinates, and multi-view geometry and interpret real-world visual scenes using OpenCV or MATLAB. (BL2)*

Assessment Tool Weightage: 40%

QUESTIONS: 20 **Total Marks:40**

Annexure A

SHORT ANSWER TEST

Code of Conduct:

I, Harini. R/ 192324108 certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:
R.Harini

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary (5 Marks)	Level 3 – Proficient (4 Marks)	Level 2 – Developing (2–3 Marks)	Level 1 – Beginning (0–1 Mark)	Marks
Conceptual Understanding	**30**	Demonstrates clear and complete understanding of the concept	Good understanding with minor gaps	Basic understanding with some misconceptions	Poor or incorrect understanding	
Accuracy of Answer	**25**	Completely correct answer with no errors	Mostly correct with minor errors	Partially correct with noticeable errors	Incorrect or irrelevant answer	
Application of Knowledge	**20**	Correctly applies concepts to solve the question/problem	Applies concepts with minor mistakes	Limited or partially correct application	No or incorrect application	
Completeness of Response	**15**	Covers all required parts of the question	Covers most parts with minor omissions	Covers only some parts	Incomplete or missing response	
Clarity & Presentation	**10**	Clear, concise, and well-organized answer	Understandable with minor issues	Somewhat unclear or poorly structured	Difficult to understand	

Annexure A

Short Answer Test

1. A mobile app enhances photos automatically. Identify whether this task belongs to image processing or computer vision and justify.

2. A game engine generates a 3D virtual environment. Classify this under computer graphics or vision and explain.

3. A system detects edges in an image before object recognition. Which level of vision is involved?

4. A robot identifies objects after extracting features. Which vision level is this?

5. A CCTV system identifies a person from a crowd. Which vision level is involved?

6. A camera captures a blurred image due to improper focus. What aspect of image formation is affected?

7. A smartphone camera shows distorted images. What is the need for camera calibration here?

8. A self-driving car uses two cameras to estimate distance. What concept is being used?

9. A drone captures images from different angles to build a 3D model. Which concept is applied?

10. A medical imaging system analyzes MRI scans to detect tumors. Which CV application area does this belong to?

11. A virtual try-on app overlays glasses on a user's face in real time. Which fields are involved: graphics, vision, or both?

12. A system converts a real-world 3D scene into a 2D image using a camera. What process of image formation is taking place?

13. A smartphone camera app corrects geometric distortion in captured images. Which concept is applied here?

14. A surveillance system detects motion but cannot identify objects. Which level of vision is represented?

15. A robot identifies objects after analyzing their shape and features. Which level of vision is involved?

16. A VR system creates a virtual environment without using real-world images. Is this computer graphics or vision? Justify.

17. A stereo camera system generates depth maps for a scene. Which concept is being applied?
18. A system analyzes multiple images of a building to estimate its structure. Which concept is used here?

19. A biometric system uses facial features to verify identity. Which application domain of computer vision does this belong to?

20. A camera fails to measure accurate depth due to improper alignment of lenses. Which concept needs correction?

DSA0201 - COMPUTER VISION WITH OPENCV
CO1 - AT1

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SHORT ANSWER TEST

1. A mobile app enhances photos automatically. Identify whether this task belongs to image processing or computer vision and justify.

This task belongs to image processing. Image processing involves modifying or improving an existing image to enhance its visual quality. In this case, the mobile application adjusts properties such as brightness, contrast, sharpness, and color balance to make the photograph look better. It does not necessarily need to identify or understand the objects present in the image.

2. A game engine generates a 3D virtual environment. Classify this under computer graphics or vision and explain.

This task belongs to computer graphics. Computer graphics is the process of creating images, animations, and three-dimensional environments using computers. A game engine generates virtual objects, textures, lighting, shadows, and backgrounds to create a realistic gaming environment. Unlike computer vision, which analyzes images captured from the real world, computer graphics generates new visual content.

3. A system detects edges in an image before object recognition. Which level of vision is involved?

This involves low-level vision. Low-level vision deals with basic image processing operations such as edge detection, noise removal, filtering, and intensity analysis. Edge detection identifies sudden changes in image intensity, which often correspond to object boundaries. These edges provide important information for later stages of object recognition.

4. A robot identifies objects after extracting features. Which vision level is this?

This involves mid-level and high-level vision. Mid-level vision extracts meaningful features such as edges, corners, shapes, and textures from an image. High-level vision uses these features to recognize, classify, and understand objects. For example, a robot may extract the shape and texture of a cup and then identify it as a cup.

5. A CCTV system identifies a person from a crowd. Which vision level is involved?

This involves high-level vision because the system interprets visual information to recognize or identify a person. It may analyze facial features, body appearance, or other distinguishing characteristics. The extracted features are compared with available reference information to identify the person. This is an example of applying computer vision to surveillance and person recognition.

6. A camera captures a blurred image due to improper focus. What aspect of image formation is affected?

The focusing process and image sharpness are affected. During image formation, light reflected from an object passes through the camera lens and should converge appropriately on the image sensor. If the lens is not focused correctly, the light from a point on the object spreads over a larger sensor area, producing a blurred image. Proper focusing helps produce a sharp and clear image.

7. A smartphone camera shows distorted images. What is the need for camera calibration here?

Camera calibration is needed to determine the camera's internal parameters and correct geometric distortions. Intrinsic parameters include focal length, the optical center, and pixel scaling, while distortion coefficients describe lens-related distortions. Calibration helps correct effects such as barrel distortion and pincushion distortion, which can make straight lines appear curved. This improves the geometric accuracy of images and measurements.

8. A self-driving car uses two cameras to estimate distance. What concept is being used?

The concept used is stereo vision. Stereo vision estimates the depth and distance of objects by comparing images captured by two cameras positioned at different locations. The system identifies corresponding points in both images and calculates their disparity, which is the difference in their image positions. Objects with greater disparity are generally closer to the cameras, while objects with smaller disparity are farther away, assuming a calibrated stereo setup.

9. A drone captures images from different angles to build a 3D model. Which concept is applied?

The concept applied is Structure from Motion (SfM), which is a multi-view reconstruction technique. It analyzes overlapping images captured from different camera positions to identify corresponding features. Using these correspondences, it estimates camera positions and

reconstructs the three-dimensional structure of the scene. This technique is useful in drone mapping, surveying, architecture, and 3D modeling.

10. A medical imaging system analyzes MRI scans to detect tumors. Which CV application area does this belong to?

This belongs to medical image analysis, an important application of computer vision. Computer vision techniques process and analyze MRI images to identify abnormal regions, tissue boundaries, and possible tumors. Image segmentation can separate suspicious regions from surrounding tissues, while classification techniques can help categorize detected abnormalities. These methods assist medical professionals in diagnosis and treatment planning.

11. A virtual try-on app overlays glasses on a user's face in real time. Which fields are involved: graphics, vision, or both?

This application involves both computer vision and computer graphics. Computer vision detects the user's face, identifies facial landmarks such as the eyes and nose, and tracks head movements in real time. Computer graphics renders the virtual glasses and adjusts their position, orientation, size, and appearance to match the user's face. Combining these fields creates an augmented reality experience in which the glasses appear to fit naturally.

12. A system converts a real-world 3D scene into a 2D image using a camera. What process of image formation is taking place?

The process is called perspective projection. A camera projects three-dimensional points from the real world onto a two-dimensional image plane through its optical center. The projected position of each point depends on its three-dimensional coordinates and the camera's intrinsic and extrinsic parameters. In perspective projection, distant objects generally appear smaller than nearby objects, creating a sense of depth in the image.

13. A smartphone camera app corrects geometric distortion in captured images. Which concept is applied here?

The concept applied is camera calibration and lens distortion correction. Camera calibration estimates the camera's intrinsic parameters and lens distortion coefficients using known points or calibration patterns. These parameters are then used to transform distorted image coordinates into corrected coordinates. For example, barrel distortion, in which straight lines bend outward near the edges, can be reduced to produce a more geometrically accurate image.

14. A surveillance system detects motion but cannot identify objects. Which level of vision is represented?

This primarily represents low-level vision when motion is detected through basic frame differences or changes in pixel intensity. The system compares consecutive video frames to identify regions where pixels have changed, indicating possible movement. However, it does not interpret the moving regions as specific objects such as people, vehicles, or animals. Object identification would require additional feature extraction and higher-level recognition.

15. A robot identifies objects after analyzing their shape and features. Which level of vision is involved?

This involves high-level vision, supported by mid-level feature extraction. The system first extracts features such as edges, corners, shapes, and textures from the captured image. It then uses these features to recognize and classify objects based on their visual characteristics. For example, a robot can distinguish a bottle from a box by analyzing their shapes and other identifying features.

16. A VR system creates a virtual environment without using real-world images. Is this computer graphics or vision? Justify.

This belongs to computer graphics. Computer graphics generates virtual scenes, objects, textures, lighting, and animations using mathematical models and rendering techniques. A virtual reality system uses these techniques to create an immersive environment that users can explore. Since the environment is generated by the computer rather than interpreted from captured real-world images, the primary field involved is computer graphics.

17. A stereo camera system generates depth maps for a scene. Which concept is being applied?

The concept applied is stereo vision-based depth estimation. A stereo camera captures two images of the same scene from slightly different viewpoints. The system finds corresponding points in the left and right images and calculates their disparity. Using the disparity, focal length, and distance between the cameras, it estimates depth and creates a depth map, in which each pixel represents the estimated distance of a scene point from the camera.

18. A system analyzes multiple images of a building to estimate its structure. Which concept is used here?

The concept used is multi-view geometry. Multi-view geometry studies the relationships between images of the same scene captured from different viewpoints. By identifying corresponding features across the images and using the camera geometry, the system estimates the three-dimensional structure of the building. Techniques such as Structure from Motion and multi-view stereo can be used to reconstruct the building's shape and dimensions.

19. A biometric system uses facial features to verify identity. Which application domain of computer vision does this belong to?

This belongs to biometric recognition, specifically face recognition and verification. The system detects a face, extracts distinctive facial features, and converts them into a representation that can be compared with a stored reference. If the similarity satisfies a specified threshold, the system verifies that the person matches the claimed identity. Applications include smartphone unlocking, access control, and identity verification systems.

20. A camera fails to measure accurate depth due to improper alignment of lenses. Which concept needs correction?

The concept that needs correction is stereo camera calibration and image rectification. In a stereo camera system, the relative position and orientation of the two cameras must be accurately calibrated. Improper alignment can make it difficult to identify corresponding points and calculate disparity correctly. Stereo rectification aligns corresponding image points along the same image rows, improving disparity calculation and enabling more accurate depth estimation.